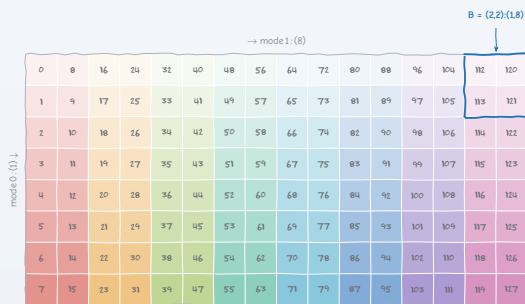


LEETCUDA

CUDA KERNEL 优化之路

从编程模型到 CUTE 与 FLASHATTENTION



L = (8,16)>(1,8)

```
auto tiled_mma = make_tiled_mma[SM80_16x8x16_F32F16F16F32_TN](); Layout<Shape<2,2,1>>(){};
```

郑重声明
本书仅用于开源学习与技术交流。
未经作者本人同意，不得用于任何商业用途。

FA2 / FA3 ATTENTION VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM-120A) B=1, H=32, N=16384, D=128

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUDNN SDPA BASELINE	232.4	1.00X	
■ FA2 MMA			
SK=2, F16ACC	157.9	0.68X	
SK=1, F32ACC	145.6	0.63X	
SK=2, F32ACC	166.5	0.72X	
■ FA2 CUTE MMA			
SK=1, F32ACC	189.6	0.82X	
SK=2, F32ACC	197.0	0.85X	
■ FA2 TMA WS (1 WG)			
SK=1, SV=1, F16ACC	161.2	0.69X	
SK=2, SV=1, F16ACC	189.4	0.81X	
SK=2, SV=2, F16ACC	190.5	0.82X	
SK=2, SV=1, F32ACC	204.9	0.88X	
■ FA3 TMA WS (2 WG)			
SK=1, SV=1, F16ACC	210.1	0.90X	
SK=1, SV=1, F32ACC	210.8	0.91X	
■ FA2 CUTE TMA WS (1 WG)			
SK=2, SV=1, F32ACC	220.0	0.95X	
SK=3, SV=1, F32ACC	223.4	0.96X	
■ FA2 CUTE TMA WS PERSISTENT-CTA (1 WG)			
D=128	242.5	1.04X	

HGEMM CUTE SWIZZLE VS CUBLAS

NVIDIA RTX PRO 5000 (SM-120A) M=N=K=4096, F16ACC; FP8 4096/8192

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUBLAS BASELINE	236.9	1.00X	
■ HGEMM CUTE SWIZZLE (F16ACC)			
S=2, BLK-SW=0	229.5	0.97X	
S=2, BLK-SW=1	231.1	0.98X	
S=3, BLK-SW=0	245.6	1.04X	
S=3, BLK-SW=1	246.7	1.04X	
■ FP8 CUTE GEMM (F32ACC)			
WS S2 128x256, 4096 ³	443.7	1.87X	
WS S2 128x256, 8192 ³	487.8	2.06X	

SPEEDUP IS RELATIVE TO THE VENDOR BASELINE
(CUBLAS FOR HGEMM, CUDNN SDPA FOR ATTENTION).

FA SPLIT-D VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM-120A) B=1, H=32, N=8192, D=320

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUDNN SDPA BASELINE	69.8	1.00X	
■ SPLIT-D TMA WS			
SK=1, SV=1	86.6	1.24X	
SK=2, SV=2	139.9	2.01X	
■ SPLIT-D TMA NON-WS			
SK=2, SV=2	187.9	2.69X	
SK=2, SV=3	188.8	2.71X	
SK=3, SV=2	204.3	2.93X	

SK = K STAGES (FA), SV = V STAGES (FA), S = STAGES (HGEMM)